

3. Cycle conditions: 98C for 30s; 6x {98C for 10s; 60C for 15s; 65C for 45s}; 72C for 5m; 4C
4. SPRI library at 0.8x, wash with 80% EtOH (Higher SPRI concentrations can be used depending on the project and oligos). Elute in 30 uL EB, with serial elutions if split across multiple tubes.
PCR product = Oligo (230bp) + F primer (22bp addition) + R primer (77bp addition) = 329bp total library
5. Run library pool after spri on 2.2% flash gel to ensure amplification and lack of smearing
6. Quantify on Nanodrop or Qubit

MPRA Vector Assembly - mpra Δ orf library

7. Digest backbone

Component	Target ug	Vol (uL)
10x CutSmart Buffer		10
Sfil		10
pMPRAv3: Δ luc: Δ xbal	10ug	
dH ₂ O		80.00
TOTAL		100.00

8. Incubate at 50C overnight
9. SPRI at 1.5x to cleanup.
10. Run 1.2% Lonza gel of cut and uncut vector to ensure proper digestion
11. Setup gibson assembly reaction.

Component	Target ug	1x
2x NEB HiFi Assembly MM		50.00
DNA oligo pool	2.2ug	
Sfil digested - pMPRAv3: Δ luc: Δ xbal	2ug	
dH ₂ O		50.00
TOTAL		100.00

12. Incubate at 50C for 1 hr
13. SPRI purification at 1.2X and elute in 20 μ L. Store in Lo-bind tubes.
14. Test transformation to dial in CFU count: 1ul (~100ng/ul) Gibson and 50ul 10 beta electrocompetent cells electroporation (2kV, 200 ohm, 25 μ F), add 950 SOC. Incubate for 1 hour at 37C while rotating. Dilute 1:100, 1:1000, 1:10000, 1:100000. Calculate CFUs. Increase or decrease the amount of original 1ml to use for 1 hour culture incubation according to desired CFU count.
15. Transform 1ul of ligated vector into 50 μ L 10-beta e.coli (dependent on test transformation) by electroporation (2kV, 200 ohm, 25 μ F). Setup 1x 50ul electroporations add 950 ul SOC to each. Add the appropriate amount of 1ml culture to obtain desired CFU count.
The typical MPRA library uses ~200-300 barcodes per unique oligo sequence tested.
(CFU = 200 x # of oligos)
16. Split into ten 1 mL aliquots of SOC, rotate tubes for 1 hour at 37°C. Independently expand in 20 mL of LB supplemented with 100 μ g/mL of carbenicillin in shaker at 37°C for 6.5 hours.
17. Serial dilute and create CFU counting plates from initial SOC split to determine library complexity/size
18. Pool aliquots to reach desired CFU complexity and purify with Qiagen Plasmid Plus maxi/midi kit.
19. Pick 16 colonies for colony PCR and Sanger sequencing with primers #1/#2 to confirm insert.

Illumina Sequencing for oligo-tag association & complexity check

20. PCR1: Capture oligo + BC combination from pMPRAv3:Δorf plasmid

Component	Vol (uL)
Q5 2X MM (NEBNEXT Hot-Start) Ultra II	50
TruSeq Universal Adapter (10um) #757	5.00
MPRA v3 TruSeq Amp2Sa F (10um) #197	5.00
pMPRA:Δorf Plasmid (200ng)	4.00
dH ₂ O	36.00
TOTAL	100.00

21. Cycle conditions: 98C for 30s; 5x {98C for 20s; 62C for 15s; 72C for 30s}; 72C for 2m; 4C

22. Spri at 1X and elute in 30 of EB

23. PCR2: Attach index/final illumina adapters

Component	Vol (uL)
Q5 2X MM (NEBNEXT Hot-Start) Ultra II	50
P5 index primer #___(10um)	5.00
P7 index primer #___(10um)	5.00
Eluted PCR1 product	20.00
dH ₂ O	20.00
TOTAL	100.00

24. Cycle conditions: 98C for 30s; 5x {98C for 10s; 62C for 15s; 72C for 30s}; 72C for 2m; 4C

25. Spri at 1X and elute in 30 of EB

26. Tapestation to quantify and pool based on molarity

27. Sequence on Illumina using 2x150+bp.

MPRA Vector Assembly - mpraΔgfp library

28. Linearize library with AsiSI

Component	Target ug	Vol (uL)
10x CutSmart Buffer		40.00
AsiSI	100 units	10.00
pMPRAv3:Δluc:Δxbal w/ library	10ug	
dH ₂ O		350.00
TOTAL		400.00

29. Incubate at 37C overnight

30. Purify in NEB Monarch Columns (5ug capacity). Use the appropriate number of columns.

Elute in 30ul EB one column at a time using elution from column one to elute column two.

31. Generate GFP amplicon using PCR (GFP:amp1)

Component	1x
Q5 2X MM (NEBNEXT Hot-Start)	25.00
primer #200	2.50
primer #201	2.50
pMPRAv3:minP-GFP (0.1 ng/uL)	1.00
dH ₂ O	19.00
TOTAL	50.00

32. Cycle conditions: 98C for 30s; 20x {98C for 10s; 60C for 15s; 72C for 45s}; 72C for 5m; 4C
 33. Add Dpn1 digestion to PCR mixture and incubate for 30 min at 37C
 34. Double SPRI with .5x to remove bigger band, then 1.5x spri, elute in 40ul
 35. 2nd PCR of GFP amplicon (GFP:amp2)

Component	1x	32x
Q5 2X MM (NEBNEXT Hot-Start)	25.00	800
primer #200	2.50	80
primer #201	2.50	80
GFP:amp1 PCR diluted 1:100	0.40	12.8
dH ₂ O	19.60	627.2
TOTAL	50.00	

36. Cycle conditions: 98C for 30s; 20x {98C for 10s; 60C for 15s; 72C for 45s}; 72C for 5m; 4C
 37. Pool reactions, 1.5x SPRI, serial elute in 100 ul
 38. Purify in Qiagen Qiaquick (10ug capacity). Use appropriate number of columns. Serial elute columns with 100 ul EB.
 39. Insert GFP orf using Gibson Assembly

Component	Target ug	1x
2x Gibson Master Mix		125.00
GFP amplicon	5.28 ug	
digested pMPRAv3:Δluc:Δxbal w/ library	1.6 ug	
dH ₂ O		125.00
TOTAL		250.00

40. Incubate at 50C for 90 min
 41. 1.5x SPRI purify elute in 40 ul
 42. Run Gibson on 1.2% Lonza Flash gel to ensure there is a band above 4kb
 43. Re-digest to remove uncut vector

Component	Target ug	1x
10x Buffer 4	1x	10.00
AsiSI	50 units	2.50
RecBCD	5 units	0.50
BSA	10ug	1.00
ATP	1mM	
pMPRAv3:Δluc:Δxbal w/ library	all gibbon product	
dH ₂ O		86.00
TOTAL		100.00

44. Incubate at 37C overnight.
 45. 1.5x SPRI purify elute in 40 ul
 46. At this stage you can perform a test transformation to evaluate the efficiency of the GFP insertion or proceed directly to the full scale library prep. Test transformation to dial in CFU count: 1ul (~100ng/ul) Gibson and 50ul 10 beta electrocompetent cells electroporation (2kV, 200 ohm, 25 μF), add 950 SOC. Incubate for 1 hour at 37C while rotating. Dilute 1:100, 1:1000, 1:10000, 1:100000. Calculate CFUs. Increase or decrease the amount of original 1ml to use for 1 hour culture incubation according to desired CFU count.

Full Scale MPRA Plasmid Preparation

47. Electroporate 10 μ l (2kV, 200 ohm, 25 μ F) into 250 μ L 10-beta cells or use amounts of Gibson and cells discovered in test transformation
48. Immediately split electroporated bacteria across 6 tubes and recover each in 2 mL of SOC for 1 hour at 37°C
49. After recovery each tube is independently added to 500 mL of TB with 100 μ g/mL of carbenicillin and grown for 16 hours at 30°C prior (3 L total)
Create CFU counting plates for 4 of the 500 mL cultures to evaluate transformation efficiency
50. After 16 hours of growth spin cultures and process pellet using the Qiagen Plasmid Plus Giga protocol or the Macherey-Nagel NucleoBond RS purification
51. Pick 16 colonies from CFU plate for colony PCR and Sanger sequencing with primers #1/#2 to confirm insert.
52. Qiagen Plasmid Plus Giga Prep to extract plasmid from bacteria. Do not exceed 7.5g bacterial cell pellet weight per column. Also, do not exceed -300mBar vacuum pressure on any of the vacuum steps or you will get bacterial genomic DNA contamination in your library.
53. Run the purified product on a gel to confirm plasmid quality. If genomic or RNA contamination is observed, repurify the plasmid a second time using a Qiagen Plasmid Plus Giga column.

MPRA RNA Extraction - Maxi Scale

Introduction

Standard MPRA protocol for RNA Extraction and GFP Capture from 1×10^8 - 5×10^8 cells. If greater than 5×10^8 cells need to be processed split cells across multiple RNeasy columns and pool prior to GFP capture.

Materials

Primers/Oligos

- 33 uM Primer #120: CCTCGATGTTGTGGCGGGTCTTGAAGTTCACCTTG/3BioTEG/
- 33 uM Primer #123: CCAGGATGTTGCCGTCCTCCTTGAAGTCGATGCCC/3BioTEG/
- 33 uM Primer #126: CGCCGTAGGTGAAGGTGGTCACGAGGGTGGGCCAG/3BioTEG/

Other

- Qiagen RNeasy Maxi Kit (Qiagen #75162)
- Molecular Biology Grade Ethanol (Fisher #BP2818500)
- RNase-free Water (Life Tech #10977023 or AM9937)
- SUPERase-In (Life Tech #AM2696)
- TURBO DNase (Life Tech #AM2239)
- 10% SDS Solution (Life Tech #15553027)
- 0.5M EDTA (Life Tech #AM9260G)
- 20X SSC (Life Tech #15557044)
- Deionized Formamide (VWR #EM-4650)
- GFP 3 Probe Mix (100 uM total)
- Sera Mag Beads 1EA MG-SA4 1UM 1% SLD 100ML (Fisher #09 981 131)
- Bead Buffer WashA
 - 0.1 M NaOH
 - 0.05 M NaCl
- Bead Buffer WashB
 - 0.1 M NaCl
- SuperScript III First-Strand Synthesis SuperMix (Life Tech #18080400)
- First Strand Gene Specific Primer
- 20 uM Primer #19: CCGACTAGCTTGGCCGC
- Agencourt RNAClean XP (Beckman #A63987)
- Agencourt AMPure XP (Beckman #A63881)
- Eppendorf DNA/RNA LoBind Tubes 1.5 mL (#22431021)
- Eppendorf DNA LoBind Tube 5.0 mL (#30108310)

Procedure

RNA Extraction

Notes

Add 4 volumes of EtOH (96-100%) to Buffer RPE before using.
Skip steps 1-3 if RLT/DTT and lysing was performed prior to freezing.
Rinse cells before harvest 3 times with PBS to remove DNA plasmid library

1. Remove frozen cell pellet from -80C and immediately add 7.5/15 mL Qiagen RLT Buffer and 150/300 μ L 2M DTT ($<2 \times 10^8$ / $>2 \times 10^8$ cells).
 - If planning to store >1 month, lyse in RLT/DTT and freeze lysate at -80C.
2. Dip tubes in 37 degree bath for 30 sec to defrost the pellet or defrost on benchtop. Homogenize lysate for 60 seconds.
3. Add 7.5/15ml (15 for $>2 \times 10^8$ cells) mL of 70% EtOH to the cell lysis

4. Transfer 15 mL of lysis/EtOH mixture to an RNeasy Maxi column and spin at 4000 x g for 5 min at 20C and discard flow through. (For $>2 \times 10^8$ cells add remaining lysate to the RNeasy Maxi column and spin at 4000 x g for 5 min and discard flow through.)
5. Add 7.5 mL of Qiagen RW1 buffer to the column and spin at 4000 x g for 5 min
6. Mix 30 μ L of Qiagen DNase and 210 μ L of Qiagen RDD buffer. Flick gently and centrifuge to collect the liquid. (x5 for five replicates)
7. Add DNase mixture to each column dropwise evenly over the entire filter.
8. Incubate for 15 minutes.
9. Add 7.5 mL of Qiagen RW1 buffer to the column and spin at 4000 x g for 5 min, discard flow through
10. Add 10 mL of Qiagen RPE buffer to the column and spin at 4000 x g for 2 min, discard flow through
11. Add 10 mL of Qiagen RPE buffer to the column and spin at 4000 x g for 2 min, discard flow through
12. Spin empty column at 4000 x g for 10 min
13. Transfer to clean collection tube. Add 750 μ L of water to the column, let sit for 2 min and spin at 4000 x g for 2 min.
14. Add 900 μ L of water to the column, let sit for 3 min and spin at 4000 x g for 7-10 min Combine elutions, add 5 μ L of SUPERase $\cdot$ In
15. Optional: Use 1 μ L for rtPCR to check for GFP transcripts

Pre-capture DNase Treatment

16. Setup DNase reaction

Component	Vol (μ L)
Total RNA	1475
Turbo DNase 10x Buffer	165
Turbo DNase	10

17. Incubate for 1 hour at 37C
18. Post incubation add the following:
 - 15 μ L of 10 % SDS
 - 150 μ L of 0.5M EDTA
19. Incubate for 5 minutes at 70C then immediately place on ice

GFP Transcript Pulldown

20. Combine the following in a 5 mL Low-Bind tube:

Component	Vol (μ L)
DNase treated RNA	1815
SSC 20X	600
Formamide	1200
GFP 3 Probe Mix - 100 μ M	2

21. Mix tube with vortex.
22. Incubate for 2.5 hours at 65C in the rotating incubator.
23. Prepare capture beads during Incubation:
 - In a 1.5 mL Low-Bind Tube wash 400 μ L (multiply by the # of replicates) of Fisher Sera Mag beads
 - Wash twice with WashA Buffer (0.1 M NaOH, 0.05 M NaCl) with 5 minute incubations

- 1x wash with WashB Buffer (0.1 M NaCl)
 - Elute beads with 500 μ l (multiply by the # of replicates) of 20x SSC
 - Add 500 μ l of RNase clean Sera Mag beads in 20x SSC to samples in 5 mL tube
24. Mix on rotator at room temperature for 15 minutes
 25. Place 5 mL tube on a magnet and remove supernatant
 26. Wash with 500 μ l 1x SSC and transfer to a 1.5 mL tube
 27. Place 1.5 mL tube on a magnet and remove supernatant
 28. Wash with 500 μ l 0.1x SSC
 29. Wash with 500 μ l 0.1x SSC
 30. Add 50 μ l of dH₂O
 31. Add 1 μ l of SUPERase•In to the ~50 μ l elution and proceed to the DNase treatment. Leave the beads in the reaction mix.

Post-capture DNase Treatment

32. Combine the following:

Component	Vol (μ L)
Total RNA + water	53.5
Turbo DNase 10x Buffer	5.5
Turbo DNase	1

33. Incubate for O/N hour at 37C (To recover more GFP mRNA see other elution procedure with 1 hour Post-capture DNase for ~1.5 Ct less on qPCR but this may occasionally result in plasmid library contamination in GFP mRNA. This procedure starts at step 44)
34. Post incubation add the following:
 - 1 μ L of 10 % SDS
35. Purify with 2x RNA Ampure XP
36. Elute RNA in 37 μ L of dH₂O
37. Add 1 μ l of SUPERase•In

GFP Transcript Reverse Transcriptase

38. Combine the following for primer annealing

Component	Vol (μ L)
DNase treated GFP mRNA	33
SuperScript III Annealing Buffer	5
Primer #19 - 20 μ M	2

39. Incubate for 5 minutes at 65C to a 4C hold. Keep on ice after incubation
40. Add the following to the reaction on ice:
 - 50 μ L of SuperScript III 2X First-Strand Reaction Mix
 - 10 μ L of SuperScript III Enzyme Mix
41. Incubate for 80 minutes at 47°, followed by 5 minutes at 85°
42. Cleanup with 2x Ampure XP
43. Elute cDNA in 30 μ l EB

GFP Transcript Pulldown-Alternative to steps 20-37 to increase amount of GFP mRNA but there is a small chance for plasmid contamination.

44. Combine the following in a 5 mL Low-Bind tube:

Component	Vol (uL)
DNase treated RNA	181500
SSC 20X	600
Formamide	1200
GFP 3 Probe Mix - 100 uM	2

45. Mix tube with vortex. Vortex for at least 15 seconds

46. Incubate for 2.5 hours at 65C in the rotating incubator.

47. Prepare capture beads during Incubation:

- In a 1.5 mL Low-Bind Tube wash 400 µl (multiply by the # of replicates) of Fisher Sera Mag Life Technologies C1 beads
- Wash twice 2x wash with WashA Buffer (0.1 M NaOH, 0.05 M NaCl) with 5 minute incubations
- 1x wash with WashB Buffer (0.1 M NaCl)
- Elute beads with 500 µl (multiply by the # of replicates) of 20x SSC
- Add 500 µl of RNase clean Sera MagC1 beads in 20x SSC to samples in 5 mL tube

48. Mix on rotator at room temperature for 15 minutes

49. Place 5 mL tube on a magnet and remove supernatant

50. Wash with 500 µl 1x SSC and transfer to a 1.5 mL tube

51. Place 1.5 mL tube on a magnet and remove supernatant

52. Wash with 500 µl 0.1x SSC

53. Wash with 500 µl 0.1x SSC

54. Add 25ul of dH₂O

55. Heat for 2 minutes at 70C, immediately place tube on a magnet

56. Remove elution and add an additional 20ul of dH₂O to the beads

57. Heat on a thermal cycler for 2 minutes at 80C, immediately place tube on magnet

58. Remove elution and combine both

59. Add 1ul SUPERase[•]In to the ~50ul elution and store at -80C

Post-capture DNase Treatment

60. Combine the following:

Component	Vol (uL)
Total RNA + water	44
Turbo DNase 10x Buffer	5
Turbo DNase	1

61. Incubate for 1 hour at 37C

62. Post incubation add the following:

1 uL of 10 % SDS

63. Purify with 2x RNA Ampure XP

64. Elute RNA in 37 uL of dH₂O

65. Add 1 ul of SUPERase[•]In

66. Proceed to GFP Transcript Reverse Transcriptase

Tag-seq Library Preparation for Illumina Sequencing

Introduction

Generation of Illumina sequencing libraries from GFP cDNA (non-UMI)

Materials

Primers/Oligos

- #801 - MPRA_Illumina_GFP_F_v2: ACTGGAGTTCAGACGTGTGCTCTTCCGATCTCGCCCTGAGCAAAGA*C*C
- #802 - Ilmn_P5_1stPCR_v2: ACTCTTTCCCTACACGACGCTCTTCCGAT*C*T
- Illumina P5 Index Primers: AATGATACGGCGACCACCGAGATCTACAC[8-10bp Index]ACACTCTTTCCCTACACGAC
- Illumina P7 Index Primers: CAAGCAGAAGACGGCATACGAGAT[8-10bp Index]GTGACTGGAGTTCAGACGTGTGC

Other

- cDNA generated from GFP RNA (RNA concentration is usually very low, less than 1ng/ul) from harvested transfected cells. Need at least 4 replicates.
- Final MPRA plasmid library
- GFP mRNA from harvested transfected cells
- Applied Biosystems QuantStudio
- 384 well plates (Cat# 4483285 Life Technologies)
- Optical Adhesive Film (Cat# 4311971 Life Technologies)
- SYBR gel stain (Cat# S7563 Life Technologies)
- Q5 Ultra Next (Cat# M0544S NEB)

Procedure

qPCR Setup for normalization and cycle determination

1. Serial dilute plasmid library from 1000pg to 10fg using 10 fold dilutions.
Use a minimum volume of 500 uL per dilution
Use EB or dH₂O + 0.01% SDS for dilutions.
2. Run plasmid samples, post-capture/DNase'd RNA, NTC and cDNA samples in duplicate

Component	Vol (uL)
Q5 NEBNext Ultra II 2X MM	5
Primers 801 (10 uM)	0.5
Primers 802 (10 uM)	0.5
Sybrgreen I (1:10,000)	1.66
cDNA	1
dH ₂ O	1.34
Total	10

3. Cycle conditions: 98C for 30s; 40x {98C for 10s; 62C for 15s; 72C for 30s}; ; Melt Curve Analysis
4. Analyze the results of qPCR
5. Find the CT (cycle) when the amplification curve for the cDNA qPCR just begins to take off and subtract one cycle. Use this number of cycles in the first illumina prep PCR.
- 6.

Starting Quantification	1st PCR Cycle #
750 pg	9
400 pg	10
200 pg	11
100 pg	12

Data derived from historical averages in the Tewhey lab.
Starting quantification equals the total mass of cDNA used
as input for PCR 1.

7. Illumina PCR 1 Setup

Component	Vol (uL)
Q5 NEB NEXT Ultra II 2X MM	25
Primers 801 (10 uM)	2.5
Primers 802 (10 uM)	2.5
cDNA	
dH ₂ O	
Total	50

8. Cycle conditions: 98C for 20s; cycles determined by qPCR {98C for 10s; 62C for 15s; 72C for 30s}; 72C for 2m; 4C
9. Spri at 1X and elute in 30 of EB
10. Second PCR 2 setup

Component	Vol (uL)
Q5 NEB NEXT Ultra II 2X MM	25
ILMN P5 Index Primer (10 uM)	2.5
ILMN P7 Index Primer (10 uM)	2.5
Eluted PCR product	20
dH ₂ O	0
Total	50

11. Cycle conditions: 98C for 20s; 6x {98C for 10s; 62C for 15s; 72C for 30s}; 72C for 2m; 4C
12. Spri at 1X and elute in 30 of EB
13. Qubit/Tapestation to quantify and confirm size if running multiple samples.
14. Sequence on Illumina with a 1x21bp read. Cluster at 80-90% max density with a 5-10% PhiX spike